

EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 5

This homework is due Friday, October 2, 2026, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned). as well as your IPython notebook saved as a PDF.

If you do not attach a PDF “printout” of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

1. Proving DTFS Properties

Let $x[n]$ be a p -periodic discrete-time signal with DTFS coefficients $\{X_k \mid k \in \langle p \rangle\}$, where $\langle p \rangle$ is any discrete interval of p contiguous integers.

For example, $\sum_{k \in \langle p \rangle} X_k$ may denote $\sum_{k=0}^{p-1} X_k$, or $\sum_{k=-p/2}^{p/2-1} X_k$ for even p . Throughout, $\omega_0 = \frac{2\pi}{p}$.

(a) Let $k_0 \in \mathbb{Z}$, and define the modulated signal

$$y[n] = e^{ik_0\omega_0 n} x[n].$$

Let Y_k denote the DTFS coefficients of $y[n]$. Prove that

$$Y_k = X_{k-k_0}, \quad \forall k \in \langle p \rangle.$$

This is called the **Modulation** property. It is the ‘dual’ of the **Time Shifting** property. Think about how ‘dual’ is an appropriate adjective here, intuitively and mathematically.

Hint: try writing down the analysis equation for both of Y_k and X_{k-k_0} .

Solution: By the analysis equation,

$$\begin{aligned} Y_k &= \frac{1}{p} \sum_{n \in \langle p \rangle} y[n] e^{-ik\omega_0 n} \\ &= \frac{1}{p} \sum_{n \in \langle p \rangle} e^{ik_0\omega_0 n} x[n] e^{-ik\omega_0 n} \\ &= \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-i(k-k_0)\omega_0 n} \\ &= X_{k-k_0}. \end{aligned}$$

Equivalently, substitute the synthesis equation for $x[n]$:

$$y[n] = e^{ik_0\omega_0 n} \sum_{\ell \in \langle p \rangle} X_\ell e^{i\ell\omega_0 n} = \sum_{\ell \in \langle p \rangle} X_\ell e^{i(\ell+k_0)\omega_0 n}.$$

Substituting $\ell + k_0 \rightarrow k$, each ℓ still covers a complete discrete interval of length p :

$$y[n] = \sum_{k \in \langle p \rangle} X_{k-k_0} e^{ik\omega_0 n}.$$

Matching coefficients of $e^{ik\omega_0 n}$ against the synthesis equation $y[n] = \sum_{k \in \langle p \rangle} Y_k e^{ik\omega_0 n}$ gives

$$Y_k = X_{k-k_0}.$$

- (b) Now suppose $x[n]$ is real-valued and odd, i.e. $x[n] = -x[-n]$ for all $n \in \mathbb{Z}$. Show that X_k is odd in k and purely imaginary. More formally, show that

$$X_{-k} = -X_k \quad \text{and} \quad X_k^* = -X_k, \quad \forall k \in \langle p \rangle.$$

Hint: for the first claim, try a change of variables $m = -n$ in the analysis equation for X_{-k} . For the second, recall that a complex number z is purely imaginary if and only if $z^ = -z$.*

Solution: We first show $X_{-k} = -X_k$. The analysis equation gives

$$X_{-k} = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{ik\omega_0 n}.$$

Substituting $m = -n$, each n still covers a complete discrete interval of length p , so

$$X_{-k} = \frac{1}{p} \sum_{m \in \langle p \rangle} x[-m] e^{-ik\omega_0 m}.$$

Using $x[-m] = -x[m]$,

$$X_{-k} = -\frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m} = -X_k.$$

We next show X_k is purely imaginary, i.e. $X_k^* = -X_k$. Starting from

$$X_k = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-ik\omega_0 n}$$

and using that $x[n]$ is real,

$$X_k^* = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{ik\omega_0 n}.$$

The same change of variables $m = -n$ yields

$$X_k^* = \frac{1}{p} \sum_{m \in \langle p \rangle} x[-m] e^{-ik\omega_0 m} = -\frac{1}{p} \sum_{m \in \langle p \rangle} x[m] e^{-ik\omega_0 m} = -X_k,$$

where the last two steps used $x[-m] = -x[m]$.

- (c) Now suppose $y[n]$ is a p -periodic, real-valued, even signal, i.e. $y[n] = y[-n]$ for all $n \in \mathbb{Z}$. Let $\{Y_k \mid k \in \langle p \rangle\}$ be the DTFS coefficients of $y[n]$. Show that $Y_k \in \mathbb{R}$ for all $k \in \langle p \rangle$.

Solution: Y_k is real if and only if $Y_k^* = Y_k$. Since $y[n] \in \mathbb{R}$:

$$Y_k = \frac{1}{p} \sum_{n \in \langle p \rangle} y[n] e^{-ik\omega_0 n},$$

$$Y_k^* = \frac{1}{p} \sum_{n \in \langle p \rangle} \left(y[n] e^{-ik\omega_0 n} \right)^* = \frac{1}{p} \sum_{n \in \langle p \rangle} y[n]^* e^{ik\omega_0 n} = \frac{1}{p} \sum_{n \in \langle p \rangle} y[n] e^{ik\omega_0 n},$$

Substituting $m = -n$, each n still covers a complete discrete interval of length p . Lastly, since $y[n]$ is even ($y[-m] = y[m]$):

$$Y_k^* = \frac{1}{p} \sum_{m \in \langle p \rangle} y[-m] e^{-ik\omega_0 m} = \frac{1}{p} \sum_{m \in \langle p \rangle} y[m] e^{-ik\omega_0 m} = Y_k.$$

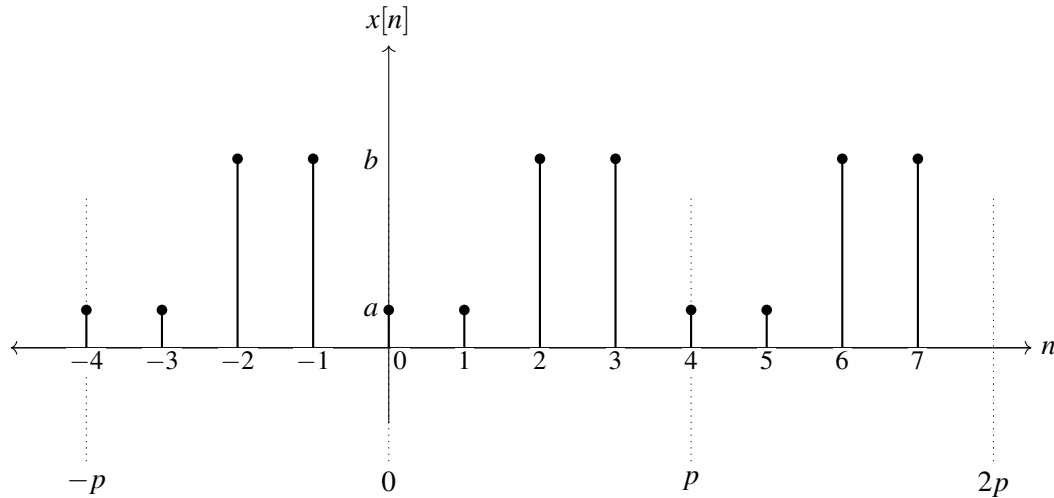
Hence $Y_k \in \mathbb{R}$.

2. “Square” Waves

Consider a p -periodic, where p is even, “square” wave signal $x[n]$. It is defined over its first period as

$$x[n] = \begin{cases} a & \text{if } 0 \leq n < p/2 \\ b & \text{if } p/2 \leq n < p \end{cases}$$

And repeats periodically. To help you understand this notation, below is a plot of $x[n]$ for $p = 4$.



- (a) Let X_k represent the DTFS coefficients of $x[n]$. Show that $X_0 = \frac{a+b}{2}$. You should prove this for a square wave with an arbitrary even period p . However, you do *not* need to compute *all* X_k for this problem.

Solution: Setting $k = 0$, the exponential in the analysis equation becomes $e^{-i0\omega_0 n} = 1$. Thus, X_0 becomes the average of the samples over one period:

$$X_0 = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] = \mu_x.$$

Over one period, $p/2$ samples equal a , and $p/2$ samples equal b , so

$$\mu_x = \frac{(p/2)a + (p/2)b}{p} = \frac{a+b}{2}.$$

Starting from the analysis equation,

$$X_0 = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n] e^{-i0\omega_0 n} = \frac{1}{p} \sum_{n \in \langle p \rangle} x[n],$$

since $e^{-i0\omega_0 n} = 1$. Because $x[n]$ is p -periodic, summing over any p contiguous integers gives the same result, so we may take $\langle p \rangle = \{0, 1, \dots, p-1\}$:

$$X_0 = \frac{1}{p} \sum_{n=0}^{p-1} x[n].$$

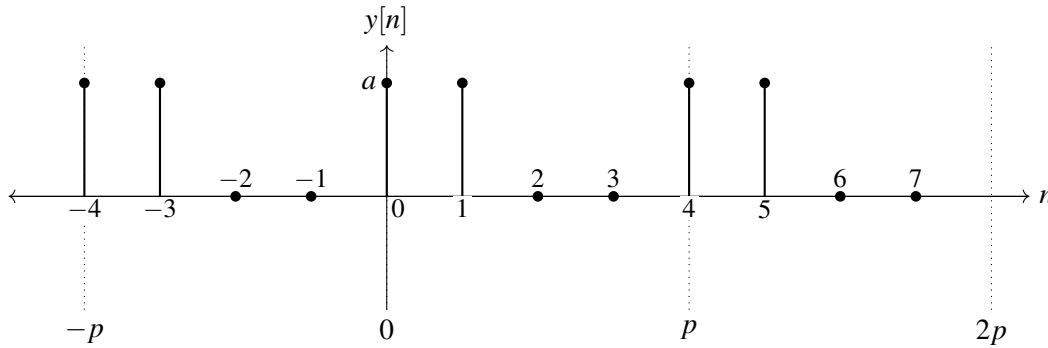
Splitting one period into its first and second halves,

$$X_0 = \frac{1}{p} \left(\sum_{n=0}^{p/2-1} x[n] + \sum_{n=p/2}^{p-1} x[n] \right).$$

Every sample in the first half ($0 \leq n \leq p/2 - 1$) equals a , and every sample in the second half ($p/2 \leq n \leq p - 1$) equals b , so each sum is just $p/2$ copies of a single value:

$$X_0 = \frac{1}{p} \left(\frac{p}{2}a + \frac{p}{2}b \right) = \frac{a+b}{2}.$$

(b) Now let consider a square wave with $b = 0$. Call this signal $y[n]$. For clarity, we plot this signal for $p = 4$:



Express the DTFS coefficient Y_k of $y[n]$ for general $k \in \langle p \rangle$. Your answer should be in terms of k , a , and p , and should include a summation.

Solution: We can apply our analysis equation in order to solve for the Fourier coefficients:

$$Y_k = \frac{1}{p} \sum_{n=0}^{p-1} y[n] e^{-ik\omega_0 n}$$

Notice that for $\frac{p}{2} \leq n \leq p - 1$, $y[n]$ is equal to 0, so those terms in the summation disappear.

$$Y_k = \frac{1}{p} \sum_{n=0}^{p/2-1} y[n] e^{-ik\omega_0 n}$$

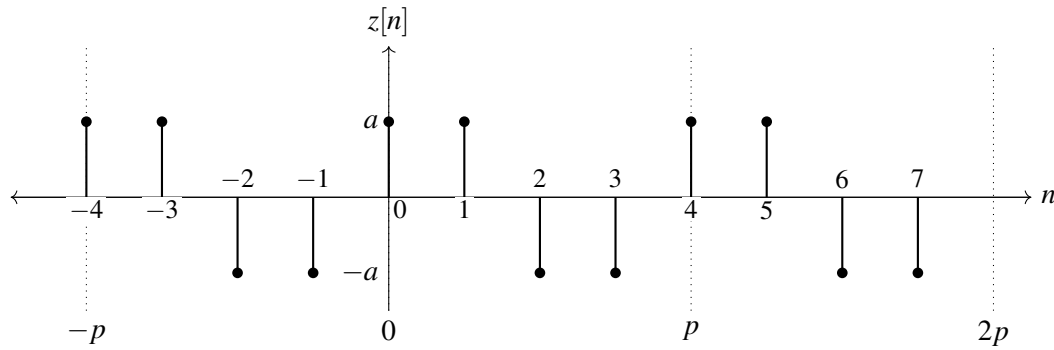
For $0 \leq n \leq \frac{p}{2} - 1$, $y[n]$ is equal to a .

$$Y_k = \frac{1}{p} \sum_{n=0}^{p/2-1} a e^{-ik\omega_0 n} = \frac{a}{p} \sum_{n=0}^{p/2-1} e^{-ik\omega_0 n}$$

For the final step, we can express ω_0 in terms of p to express Y_k in the desired terms:

$$Y_k = \frac{a}{p} \sum_{n=0}^{p/2-1} e^{-ik \frac{2\pi}{p} n}$$

(c) Now assume that $z[n]$ is a signal defined the same as $x[n]$, except the value of b is now equal to $-a$. This signal is now “half-wave symmetric.” Again, the case in which $p = 4$ is shown below:



Express the DTFS coefficients Z_k in terms of Y_k and k .

Hint: Recognize $z[n]$ as the sum of two square waves. Then, recall the time shift property: if $x[n]$ has DTFS coefficients X_k , then the shifted signal $x[n - n_0]$ has DTFS coefficients $e^{-ik\omega_0 n_0} X_k$.

Solution: Following the hint, we notice that $z[n]$ is $y[n]$ minus a copy of $y[n]$ shifted by $\frac{p}{2}$ to the right. Mathematically:

$$z[n] = y[n] - y\left[n - \frac{p}{2}\right]$$

Let $\tilde{Y}_k$ represent the DTFS coefficients of $y\left[n - \frac{p}{2}\right]$. Due to the linearity of DTFS, we know that

$$Z_k = Y_k - \tilde{Y}_k$$

Applying the time shift property with a right shift of $n_0 = p/2$:

$$\tilde{Y}_k = e^{-ik\omega_0 \frac{p}{2}} Y_k = e^{-ik \frac{2\pi}{p} \frac{p}{2}} Y_k = e^{-ik\pi} Y_k$$

$$Z_k = Y_k - e^{-ik\pi} Y_k$$

$$Z_k = Y_k (1 - e^{-ik\pi}) = Y_k (1 - (-1)^k)$$

(d) Using your result from part (c), show that $Z_k = 0$ for any even $k \in \langle p \rangle$.

Hint: any even integer k can be expressed as $k = 2c$ for another integer c .

Solution: If k is even, then $k = 2c$ for some integer c . The result from part (c) becomes

$$Z_k = Y_k (1 - e^{-i(2c)\pi}) = Y_k (1 - (e^{-i2\pi})^c) = Y_k (1 - 1^c) = Y_k (1 - 1) = 0$$

In fact, for any half-wave symmetric waveform, the even Fourier coefficients are 0; the steps we did in parts (c) and (d) didn't rely on the fact that our signal was a square wave!

(e) Another way to build $z[n]$ out of $y[n]$:

$$z[n] = 2y[n] - a.$$

Check that this visually makes sense, then write Z_k in terms of Y_k and compare with your answer to part (c). What does this tell you about Y_k ?

Hint 1: for a constant signal, the only non-zero DTFS coefficient is the $k = 0$ one.

Solution: A constant signal equal to a is just a times the $k = 0$ basis signal $e^{i(0)\omega_0 n} = 1$, so its DTFS coefficients are a at $k = 0$ and 0 for every other $k \in \langle p \rangle$. By linearity of the DTFS,

$$Z_k = 2Y_k - \begin{cases} a & k = 0 \\ 0 & k \neq 0. \end{cases}$$

Since the DTFS coefficients of $z[n]$ are unique, this must agree with part (c), $Z_k = Y_k (1 - (-1)^k)$, for every k :

- $k = 0$: $0 = 2Y_0 - a$, so $Y_0 = \frac{a}{2}$. This recovers part (a) with $b = 0$.
- k even, $k \neq 0$: $0 = 2Y_k$, so $Y_k = 0$.
- k odd: $2Y_k = 2Y_k$, which gives no new information.

So *every even coefficient of $y[n]$ other than Y_0 is zero*. Note that this is a statement about $y[n]$, not $z[n]$! Further, we obtained it without even evaluating the summation from part (b), using the fact that the DTFS decomposition is unique.

3. Mystery DTFS (Adapted from EE 120 Discussion)

Consider a periodic real-valued discrete-time signal x having fundamental period p and an exponential discrete Fourier series expansion

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad (1)$$

where ω_0 is the fundamental frequency of the signal. The signal has the following properties:

- (i) $x[n+3] = x[n], \forall n \in \mathbb{Z}$
- (ii) $\sum_{n=0}^{p-1} x[n] = 0$
- (iii) $\sum_{k=0}^{p-1} X_k = 0$
- (iv) $\sum_{k=0}^{p-1} |X_k|^2 = \frac{1}{2}$

Show that the signal x can be expressed in the form $e^{i(An+B)} \sin(Cn)$, and determine what the parameters A , B , and C are. There may be multiple possible sets of parameters that satisfy the conditions.

Each property reveals something new about the DTFS expansion or the coefficients X_k ; consider them in the order given.

Solution:

From property (i), we can see that $x[n]$ has a period of 3. Therefore, we can plug in $p = 3$ and $\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{3}$ for the rest of our properties.

Recall the analysis equation

$$X_k = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] e^{-ik\omega_0 n}$$

With our analysis equation, we can solve X_0 as

$$X_0 = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] e^{-i(0)\frac{2\pi}{3}n} = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] e^0 = \frac{1}{p_x} \sum_{n=0}^{p_x-1} x[n] = \frac{1}{p_x} (0) = 0$$

From here, we notice that

$$\sum_{k=0}^{p-1} X_k = \sum_{k=0}^{3-1} X_k = X_0 + X_1 + X_2 = 0$$

Since $X_0 = 0$, we have $X_0 + X_1 + X_2 = 0 + X_1 + X_2 = 0$, which means that $X_1 = -X_2$.

Finally, we can expand the last property as follows:

$$\begin{aligned}
 \sum_{k=0}^{p-1} |X_k|^2 &= \frac{1}{2} \\
 |X_0|^2 + |X_1|^2 + |X_2|^2 &= \frac{1}{2} \\
 |0|^2 + |-X_2|^2 + |X_2|^2 &= \frac{1}{2} \\
 0 + |X_2|^2 + |X_2|^2 &= \frac{1}{2} \\
 2|X_2|^2 &= \frac{1}{2} \\
 |X_2|^2 &= \frac{1}{4}
 \end{aligned}$$

From here there are many possible values for X_2 we could choose (an infinite number, actually), but the simplest is $X_2 = \frac{1}{2}$.

At this point, we have that $X_0 = 0$, $X_1 = -X_2 = -\frac{1}{2}$, and $X_2 = \frac{1}{2}$. Plugging these into our synthesis equation, we get:

$$\begin{aligned}
 x[n] &= \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \\
 &= \sum_{k=0}^2 X_k e^{ik\frac{2\pi}{3}n} \\
 &= X_0 e^{i(0)\frac{2\pi}{3}n} + X_1 e^{i(1)\frac{2\pi}{3}n} + X_2 e^{i(2)\frac{2\pi}{3}n} \\
 &= (0) e^{i(0)\frac{2\pi}{3}n} + \left(-\frac{1}{2}\right) e^{i(1)\frac{2\pi}{3}n} + \left(\frac{1}{2}\right) e^{i(2)\frac{2\pi}{3}n} \\
 &= -\frac{1}{2} e^{i\frac{2\pi}{3}n} + \frac{1}{2} e^{i\frac{4\pi}{3}n}
 \end{aligned}$$

To isolate a sine, we rewrite the exponents so both terms share the factor $e^{i\pi n}$, then factor it out:

$$\begin{aligned}
 &= \frac{1}{2} e^{(i\pi n + i\frac{\pi}{3}n)} - \frac{1}{2} e^{(i\pi n - i\frac{\pi}{3}n)} \\
 &= \frac{1}{2} e^{i\pi n} (e^{i\frac{\pi}{3}n} - e^{-i\frac{\pi}{3}n})
 \end{aligned}$$

This difference of exponentials is exactly Reverse Euler's formula $\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$, so we rewrite $\frac{1}{2}$ as $\frac{i}{2i}$ (which equals $\frac{1}{2}$):

$$\begin{aligned}
 &= e^{i\pi n} i \frac{(e^{i\frac{\pi}{3}n} - e^{-i\frac{\pi}{3}n})}{2i} \\
 &= e^{i\pi n} \cdot i \sin\left(\frac{\pi}{3}n\right)
 \end{aligned}$$

Since Euler's formula gives $i = e^{i\frac{\pi}{2}}$, we convert the leftover i into an exponential so all the factors combine into one phase:

$$\begin{aligned} &= e^{i\pi n} e^{i\frac{\pi}{2}} \sin\left(\frac{\pi}{3}n\right) \\ &= e^{i(\pi n + \frac{\pi}{2})} \sin\left(\frac{\pi}{3}n\right) \end{aligned}$$

Matching this against the target form $x[n] = e^{i(An+B)} \sin(Cn)$, we can see that $A = \pi$, $B = \frac{\pi}{2}$, and $C = \frac{\pi}{3}$ are parameters such that $x[n] = e^{i(An+B)} \sin(Cn)$ satisfies all of our described qualities.

4. Parseval's Theorem

Let x be a **real valued** periodic signal that repeats every p samples. Additionally, let X_k be the coefficients in the DTFS expansion of x . It turns out that, when you transform into the DTFS basis, “energy” is conserved! This is stated formally by Parseval's theorem:

$$p \cdot \sum_{k=0}^{p-1} |X_k|^2 = \sum_{n=0}^{p-1} (x[n])^2$$

In this problem, we prove this fact using the linear-algebraic perspective of DTFS.

- (a) Let $V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an orthogonal basis that spans $\mathbb{C}^n$, where each vector $\mathbf{v}_i$ has the **same norm** $\|\mathbf{v}_i\|$. In particular,

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} \|\mathbf{v}_i\|^2 & i = j, \\ 0 & i \neq j. \end{cases}$$

A vector $\mathbf{y} \in \mathbb{R}^n$ can be expressed in the V basis as

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \dots + \beta_n \mathbf{v}_n.$$

Write $\langle \mathbf{y}, \mathbf{y} \rangle$ in terms of $y_1, \dots, y_n$ **and** in terms of $\beta_1, \dots, \beta_n$.

Notice that even though $\mathbf{y}$ is real valued, the vectors $\mathbf{v}_i$ are **complex valued**. This is exactly what we do in DTFS! That is, we express a real valued signal in terms of the complex valued DTFS basis.

Solution: We use the fact that $\langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{y}\|^2 = \mathbf{y}^T \mathbf{y}$. Then, we can rewrite $\langle \mathbf{y}, \mathbf{y} \rangle$ in terms of $y_1, \dots, y_n$:

$$\langle \mathbf{y}, \mathbf{y} \rangle = \mathbf{y}^T \mathbf{y} = y_1^2 + y_2^2 + \dots + y_n^2 = \sum_{i=1}^n y_i^2$$

We can also express $\langle \mathbf{y}, \mathbf{y} \rangle$ in basis V as follows:

$$\langle \mathbf{y}, \mathbf{y} \rangle = \langle \beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n, \beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n \rangle$$

By the properties of complex inner products and the fact that $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $i \neq j$, we can simplify:

$$\langle \beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n, \beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n \rangle = \beta_1 \beta_1^* \langle \mathbf{v}_1, \mathbf{v}_1 \rangle + \dots + \beta_n \beta_n^* \langle \mathbf{v}_n, \mathbf{v}_n \rangle$$

Since $\langle \mathbf{v}_i, \mathbf{v}_i \rangle = \|\mathbf{v}_i\|^2 = C$ (since all the $\mathbf{v}_i$ s have the same norm) and $\beta_i \beta_i^* = |\beta_i|^2$, we can simplify further:

$$\beta_1 \beta_1^* \langle \mathbf{v}_1, \mathbf{v}_1 \rangle + \dots + \beta_n \beta_n^* \langle \mathbf{v}_n, \mathbf{v}_n \rangle = \|\mathbf{v}_1\|^2 |\beta_1|^2 + \dots + \|\mathbf{v}_n\|^2 |\beta_n|^2 = C \cdot \sum_{i=1}^n |\beta_i|^2$$

Thus,

$$\langle \mathbf{y}, \mathbf{y} \rangle = \sum_{i=1}^n y_i^2 = C \cdot \sum_{i=1}^n |\beta_i|^2$$

- (b) Using what we have learned above, show that:

$$p \cdot \sum_{k=0}^{p-1} |X_k|^2 = \sum_{n=0}^{p-1} (x[n])^2.$$

Hint: what is the norm of each DTFS basis vector?

Solution: Instead of dealing with vectors in $\mathbb{R}^n$ and $\mathbb{C}^n$, we have vectors in $\mathbb{R}^p$ and $\mathbb{C}^p$. This is because we can represent one period of the signal as

$$\mathbf{x} = [x[0] \quad x[1] \quad \cdots \quad x[p-1]]^T$$

and each of the DTFS basis vectors as

$$\boldsymbol{\psi}_k = [1 \quad e^{i2\pi k/p} \quad \cdots \quad e^{i2\pi k(p-1)/p}]^T.$$

These vectors are mutually orthogonal, and we know that each one has squared norm

$$\|\boldsymbol{\psi}_k\|^2 = \sum_{n=0}^{p-1} \left| e^{i2\pi kn/p} \right|^2 = p.$$

These should remind you of the $\mathbf{v}_i$ in part (a)! Convince yourself, if you have not already, that the DTFS synthesis formula is an expansion analogous to that of $\mathbf{y}$ in terms of the $\mathbf{v}_i$ in part (a).

Pattern matching and applying the result of part (a) with $\|\mathbf{v}_i\| = \sqrt{p}$ and $\beta_k = X_k$ yields

$$\langle \mathbf{x}, \mathbf{x} \rangle = p \sum_{k=0}^{p-1} |X_k|^2$$

and

$$\langle \mathbf{x}, \mathbf{x} \rangle = \sum_{n=0}^{p-1} |x[n]|^2 = \sum_{n=0}^{p-1} (x[n])^2$$

since x is real valued. Combining these two expressions proves Parseval's theorem:

$$p \sum_{k=0}^{p-1} |X_k|^2 = \sum_{n=0}^{p-1} (x[n])^2.$$

5. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

Solution:

I worked alone because my study group was out of town.

I prefer to work alone because a group can be very distracting for me.

I worked with my friends Alice and Bob. It's nice to be able to bounce ideas off one another and fill in each other's gaps. Explaining my ideas to someone else is hard, but it really makes sure I understand them well!

I met some new friends Carol and Danny at homework party, and we were able to finish most of the homework together with occasional help from a tutor. I reviewed my answers on my own to make sure I understood everything.

- (b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

Solution:

I didn't remember some of the definitions I learned in linear algebra, so I consulted my old notes and reviewed concepts as I needed to while solving the problems

I went to office hours to work on the homework, and whenever I got stuck, I asked one of the tutors to provide some hints or review concepts I didn't remember.

I tried all the problems on my own, and then asked an LLM to critique my approach. I got 90% of the way there on my own, but this interaction enriched my understanding of the material!

I reviewed my lecture notes before starting the homework; and whenever I got stuck, I scoured the discussion worksheets for similar problems and used the solutions to guide my thought process.

I asked some questions on Ed whenever I was confused so that I could get the most fitting answer to my question, in the context of EE66. I also answered some questions because this is a good way to practice the stuff I know well.

I couldn't even understand what problem 2 was asking, so I asked an AI tool to review all the relevant prerequisite knowledge and explain the problem statement in a different way before trying to tackle it on my own.

- (c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

Solution:

I found question 3 very intriguing because this is a totally different perspective on DTFS to what we learned in lecture. Question 2 was pretty boring and mechanical, because we were just practicing the synthesis equation, but I suppose it is important to be able to do these more-basic things too.

- (d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

Solution: I did feel prepared! But, question 5 seemed to me to be out of place because it's been two weeks since we covered least squares. We talked about SVD and its application for PCA in class and discussion this week, and I didn't really understand any of it, so it would have been nice to have some problems which gave me an opportunity to practice those concepts.

(e) Is there anything else you would like to share?

Solution:

Johnny, who is a tutor, was SUPER DUPER helpful in office hours. He explained stuff in such a way that I still felt like I was discovering the answer to the problems by myself, which was awesome.

I'm feeling very overwhelmed by this class and the upcoming midterm is really stressing me out. These homeworks are really hard and I wish there were more resources available, which I could use to understand them deeply.